

CHAPTER – 1A

TRAFFIC ACCIDENTS

Introduction

India has a well-knit and coordinated system of transport which plays an important role in development of economic activities by promoting fair distribution of produced goods and services and movement of people. The share of transport sector in Gross Domestic Product (GDP) of India is steadily growing. It is one of the key indicators in assessment of socio-economic development of the country. Since traffic accidents are indicator of bottlenecks and other hindrances in smooth flow of traffic, hence NCRB collects detailed data on traffic accidents including road accidents for inferring the trend and patterns of traffic accidents for the planners to devise appropriate preventive strategies.

The Bureau collects data on ‘Traffic Accidents’ comprising of (i) Road Accidents (ii) Railway Accidents and (iii) Railway Crossing Accidents, as these are the major contributors of accidental deaths.

Number of ‘Traffic Accidents’ in the country have decreased from 4,67,171 in 2019 to 3,68,828 in 2020. (However, the rate of deaths in road accidents per thousand vehicles has declined from 0.52 in 2019 to 0.45 in 2020). Maximum decrease in number of traffic accidents cases in States from 2019 to 2020 was reported in Tamil Nadu (from 59,499 to 46,443) followed by Kerala (from 40,354 to 27,998), Uttar Pradesh (from 42,368 to 30,593),

Madhya Pradesh (from 53,379 to 43,360), Maharashtra (from 33,624 to 24,908). [Table-1A.1].

These traffic accidents resulted in injuries to 3,36,248 persons and 1,46,354 deaths during 2020. State of Uttar Pradesh (21,156 deaths) followed by Maharashtra (13,940 deaths) and Madhya Pradesh (12,629 deaths) have reported maximum fatalities in traffic accidents in the country; these 3 States accounted for 14.5%, 9.5% and 8.6% of total deaths in traffic accidents respectively and collectively accounted for 32.6% (47,725 out of 1,46,354) of total fatalities reported at all India level during 2020.

The percentage share of traffic accidental deaths in total deaths due to ‘Other Causes’ has decreased from 43.4% in 2016 to 39.9% in 2020. A rising trend was seen in absolute number of deaths in ‘Traffic Accidents’ since 2016 to 2019, with sudden decline in 2020. Number of deaths have decreased by 19.2% (from 1,81,113 to 1,46,354) in 2020 over 2019 [LIST-1A.1].

These traffic accidents comprise of 3,54,796 road accidents, 13,018 railway accidents and 1,014 railway crossing accidents and caused 1,33,201, 11,968 and 1,185 deaths respectively during 2020.

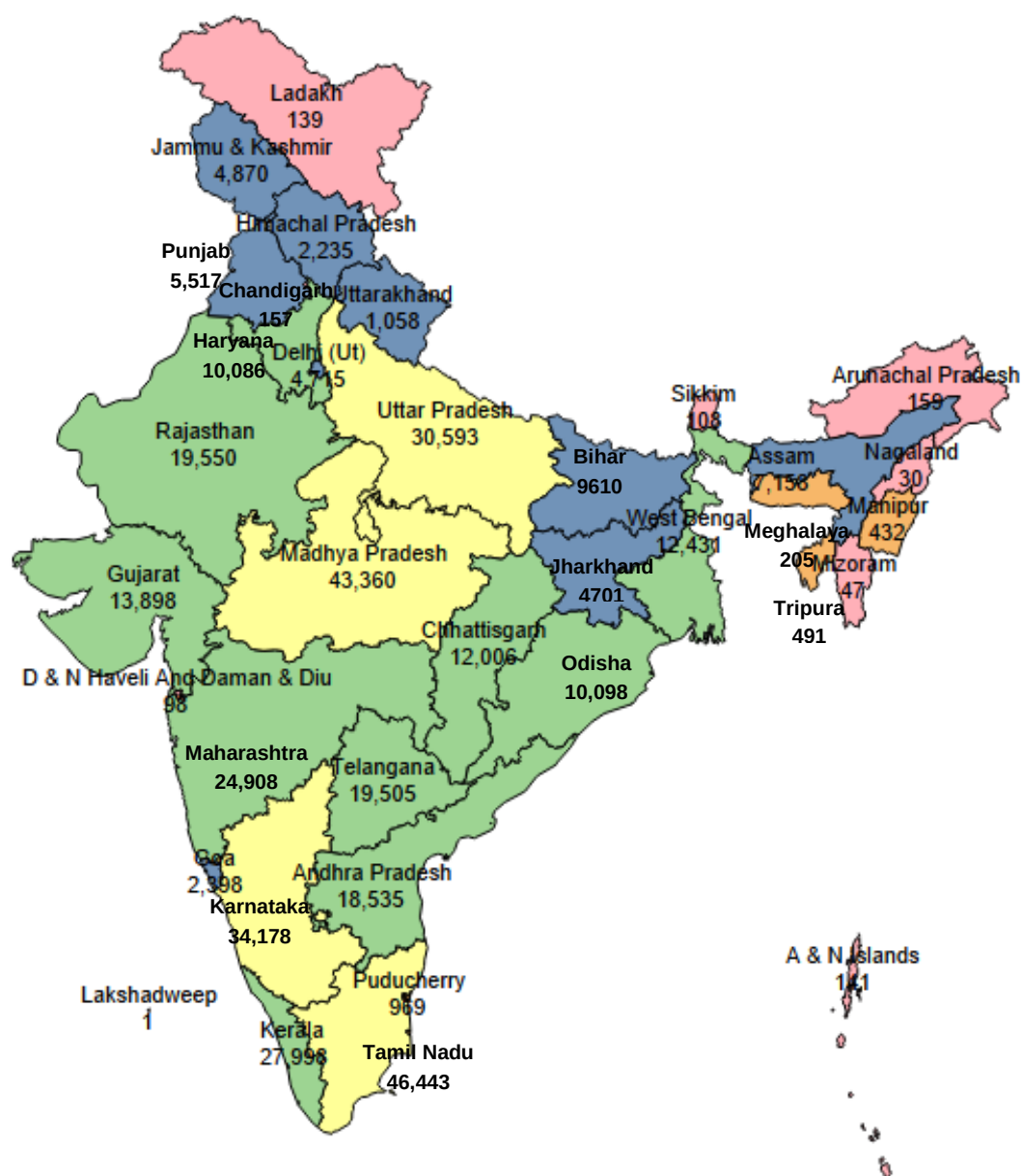
LIST-1A.1

Number and Share of Deaths due to Traffic Accidents during 2016 – 2020

Sl. No.	Year	Number of Accidental Deaths					Percentage Share of ‘Traffic Accidental Deaths’ in Accidental Deaths due to ‘Other Causes’
		Road Accidents	Railway Accidents	Railway Crossing Accidents	Total Traffic Accidents	Total Accidental Deaths due to ‘Other Causes’	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1	2016	1,51,801	22,970	3,133	1,77,904	4,09,537	43.4%
2	2017	1,50,093	23,959	1,534	1,75,586	3,89,441	45.1%
3	2018	1,52,780	24,545	1,507	1,78,832	4,04,933	44.2%
4	2019	1,54,732	24,619	1,762	1,81,113	4,12,959	43.9%
5	2020	1,33,201	11,968	1,185	1,46,354	3,66,992	39.9%

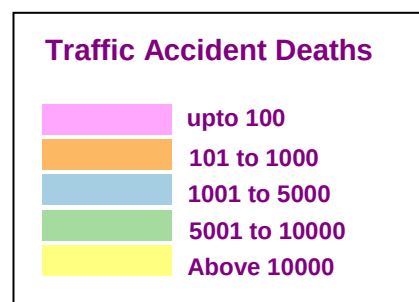
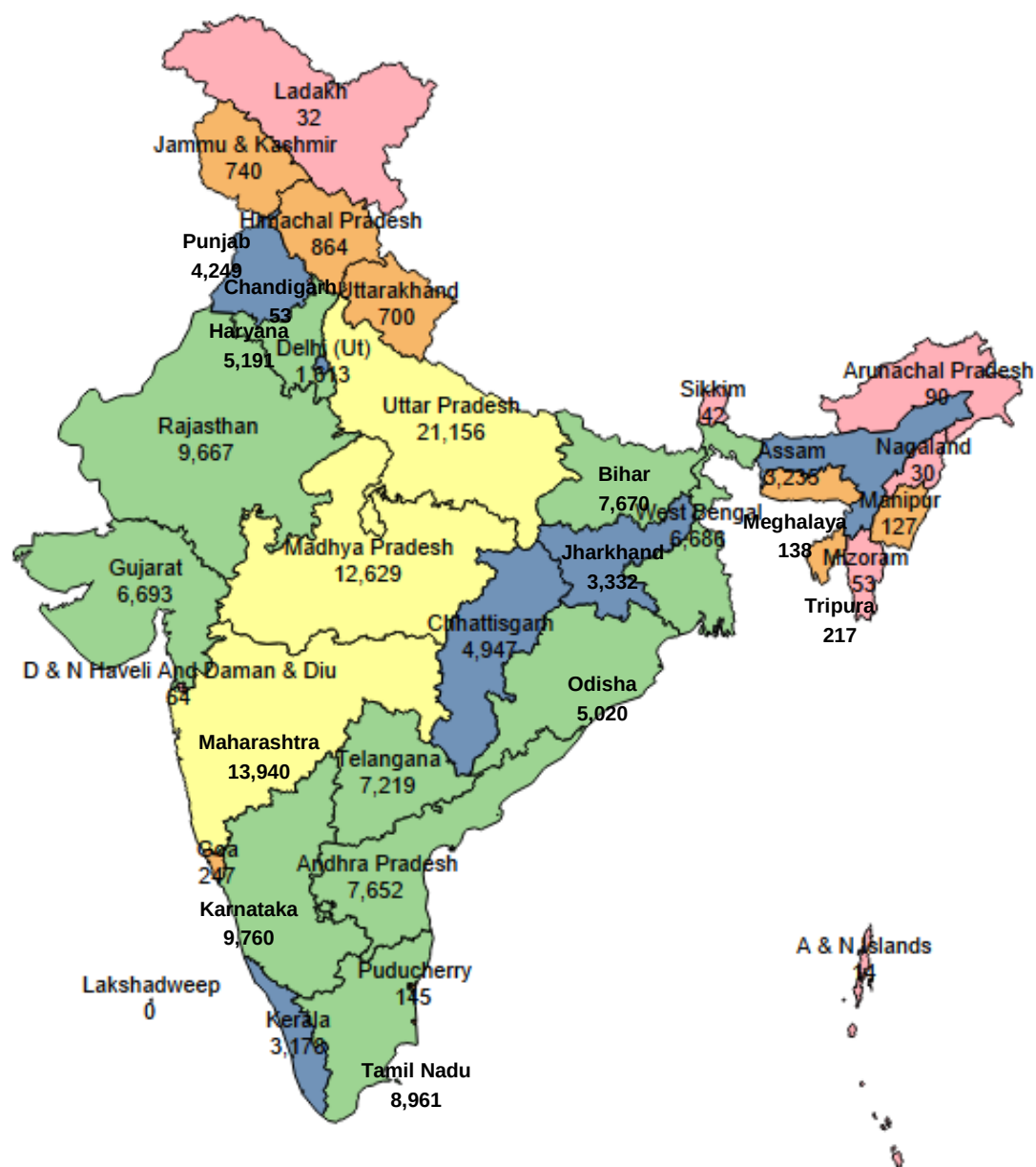
- As per data provided by States/UTs.

FIGURE-1A.1
STATE/UT – WISE TRAFFIC ACCIDENT CASES DURING 2020



• As per data provided by States/UTs.

FIGURE-1A.2
STATE/UT – WISE TRAFFIC ACCIDENT DEATHS DURING 2020



- As per data provided by States/UTs.

Month - wise patterns of traffic accidents reveal that maximum number of 'Traffic Accidents' have occurred in the month of January which accounted for 11.2%. The month-wise break-up of 'Traffic Accidents' for each State/UT/City are presented in **Table-1A.5**.

Time of occurrence - wise analysis of traffic accidents reveal that maximum number of 'Traffic Accidents' have taken place during 18:00 hrs – 21:00 hrs and 15:00 hrs – 18:00 hrs, accounting for 19.8% (70,183) and 17.7% (62,927) of total traffic accidents (3,68,828) respectively during the year 2020. State/UT-wise 'Traffic Accidents' by time of occurrence is presented in **Table-1A.6**.

Road Accidents

The Bureau has made an effort to capture comprehensive information on road accidents using the revised proforma and published the first report for the year 2014 and this edition is seventh in the series.

A total of 3,54,796 road accident cases were reported during 2020. Road accident cases in the country have decreased from

4,37,396 in 2019 to 3,54,796 in 2020 [**Table-1A.1**]. The fatalities in road accidents have decreased by 13.9% (from 1,54,732 in 2019 to 1,33,201 in 2020). The **LIST-1A.1** can be referred to see the patterns of 'Road Accidental Deaths'.

The number of vehicles, number of road accidents along with resultant fatalities and injuries therefrom, their percentage variations over previous year and the rate of accidental deaths per thousand vehicles during the last five years are presented in **LIST-1A.2**. It is observed that the rate of deaths per thousand vehicles in 2020 has declined from 0.52 in 2019 to 0.45 in 2020.

3,54,796 road accidents caused 1,33,201 deaths and injuries to 3,35,050 persons during 2020. Generally road accidents have caused more injuries than deaths, but in Mizoram, Punjab and Uttar Pradesh, road accidents caused more deaths compared to persons injured. In Mizoram, 47 road accidents caused 53 deaths & injuries to 45 persons, in Punjab, 5,173 road accidents caused 3,916 deaths and injuries to 2,881 persons and in Uttar Pradesh, 28,653 road accidents caused 19,037 deaths and injuries to 15,982 persons. [**Table-1A.2**].

LIST-1A.2

Growth in Number of Vehicles and Road Accidents in India (2016–2020)

Sl. No.	Year	Road Accidents (in thousand)	% Variation over Previous Year	Persons Injured (in thousand)	% Variation over Previous Year	Persons Killed (in Nos.)	% Variation Over Previous Year	No. of Vehicles (In Thousand)#	% Variation over previous Year	Rate of Deaths per thousand Vehicles (Col.7/Col.9)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
1	2016	473	1.80%	485.5	0.60%	1,51,801	2.10%	2,30,031	9.50%	0.66
2	2017	445.7	-5.80%	456.2	-6.00%	1,50,093	-1.10%	2,53,311	10.10%	0.59
3	2018	445.5	0.00%	446.5	-2.10%	1,52,780	1.80%	2,72,988	7.80%	0.56
4	2019	437.4	-1.8	439.2	-1.6	1,54,732	1.3	2,97,190	8.90%	0.52
5	2020	354.8	-18.9	335	-23.7	1,33,201	-13.9	2,97,190*	-	0.45+

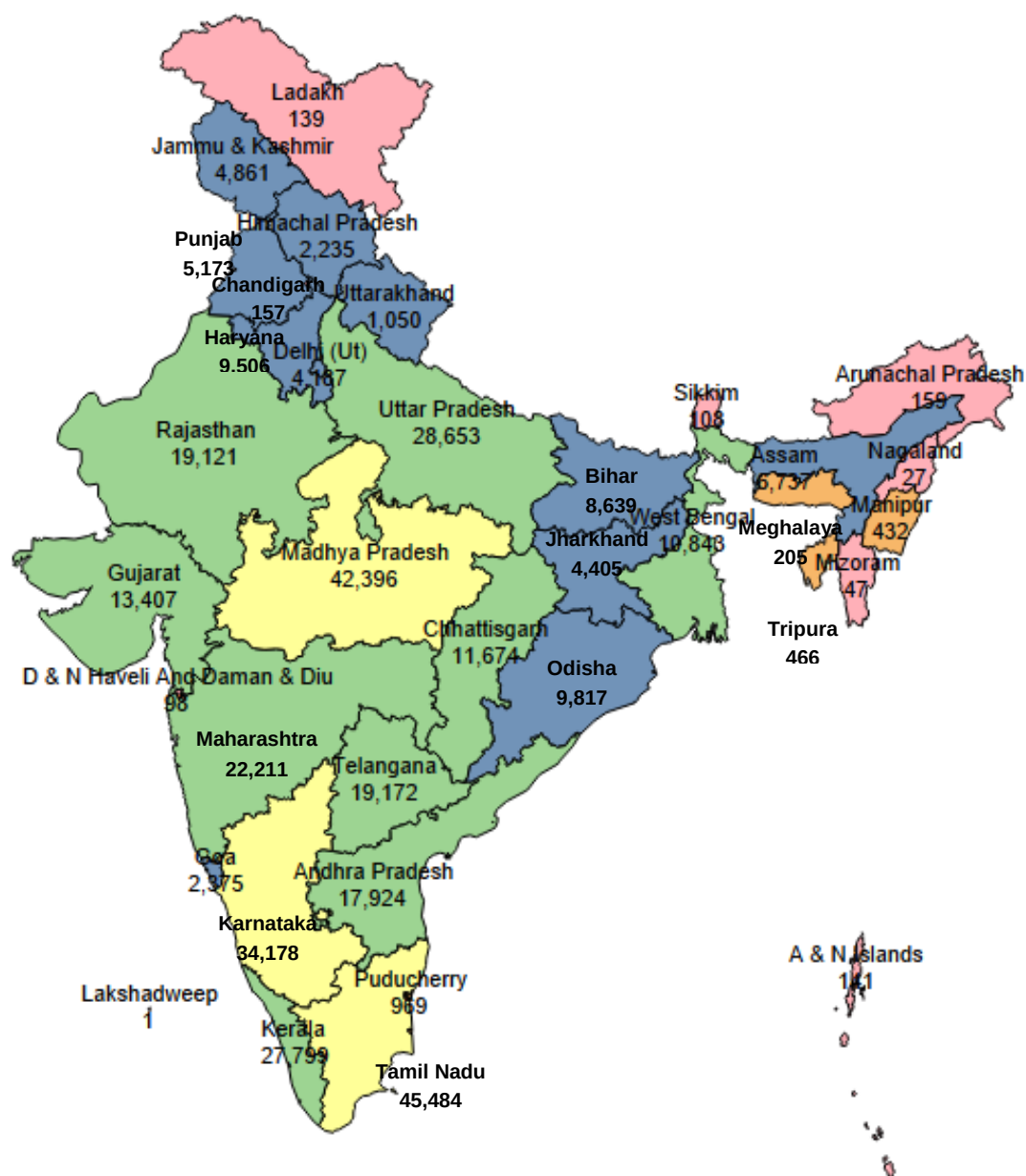
Source: Road Accidents in India - 2019, TRW, MoRT&H, as per latest published data.

* figures of the previous year used due to non-availability of data.

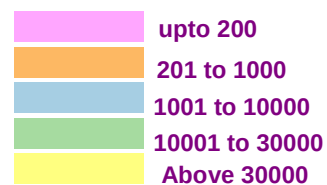
+' for calculation, figures of previous year regarding number of vehicles has been used due to unavailability of data.

- As per data provided by States/UTs.

FIGURE-1A.3
STATE/UT – WISE ROAD ACCIDENT CASES DURING 2020



Road Accident Cases



- As per data provided by States/UTs.

During 2020, two wheelers have accounted for maximum fatal road accidents (58,129 deaths), contributing 43.6% of total road accidental deaths, followed by cars (17,538 deaths) (13.2%) and trucks/lorries (16,993 deaths) (12.8%) [Table-1A.3].

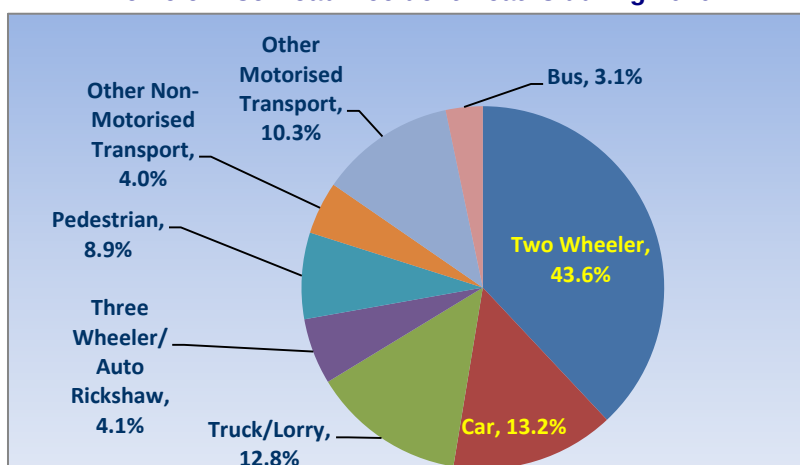
Majority of deaths due to two wheelers accidents were reported in Maharashtra (5,877 deaths) and Uttar Pradesh (5,735 deaths), accounting for 10.1% and 9.9% of total deaths due to two wheeled vehicles respectively. Large number of deaths due to car accidents (3,190 out of 17,538) were reported in Uttar Pradesh (18.2%) of total such accidents and large number of deaths due to trucks/lorries accidents (4,606 out of 16,993) were also reported in Uttar Pradesh, accounting for 27.1%. 17.7% (725 out of 4,098) and 9.8% (402

out of 4,098) of total fatal road accidents due to buses were reported in Uttar Pradesh and Bihar respectively. Deaths of 12.9% (1,536 out of 11,901) of pedestrians in road accidents were reported in Karnataka during 2020 [Table-1A.4].

The month-wise distribution of 'Road Accidents' shows that most of road accidents were reported in the month of January 38,866 cases), contributing 11.0% of total road accidents. Majority of accidents in this month (January) have been reported in Tamil Nadu, accounting for 12.0% of total accidents reported (4,675 out of 38,866 cases) in the month of January [Table-1A.5].

FIGURE-1A.4

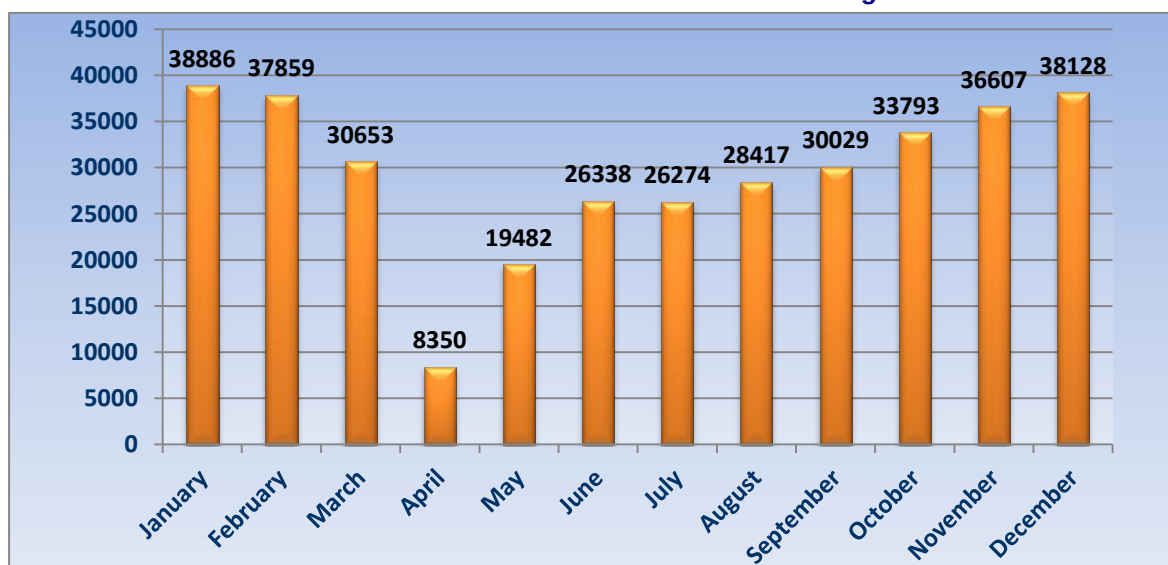
Vehicle wise Road Accident Deaths during 2020



Other Motorised Transport includes SUV/Station Wagon, Jeep, Tractor etc.

FIGURE-1A.5

Month-wise occurrence of Road Accidents during 2020



- As per data provided by States/UTs.

Most of road accidents (70,183 out of 3,54,796 cases) were reported during 18:00 hrs to 21:00 hrs (Night), accounting for 19.8% of total road accidents. During 18:00 hrs to 21:00 hrs (Night), majority of road accidents were reported in Tamil Nadu (10,751 cases), Madhya Pradesh (9,008 cases) and Karnataka (6,914 cases). Time period '15:00 hrs to 18:00 (Day)' and '12:00 hrs to 15:00 hrs (Day)' accounted for 17.7% (62,927 cases) and 15.4% (54,813 cases) of total road accidents during 2020 [Table-1A.6].

Road-wise classification of accidents is presented in Table-1A.7. As per road classification, the National Highways which has a share of only 2.0% of total road length* (1.26 Lakh KMs out of 62 Lakh KMs) accounted for highest road accidents, contributing 30.1% of total road accidents. State Highways having the share of 3.0% (1.87 Lakh KMs) of total road length have reported 24.7% of road accidents in the country. However, a considerable number of road accidents were also reported on other roads. These accounted for 45.1% of total such accidents during 2020.

A total of 1,843 cases of road accidents were also reported on Expressways which caused injuries to 1,338 persons and deaths of 1,169 persons. The highest numbers of deaths in road accidents were reported on

the National Highways accounting for 34.0% (45,275 out of 1,33,201) followed by State Highways (25.4%) (33,824 deaths). A total of 54,102 (40.6%) persons died due to road accidents on the other roads during 2020.

State/UT-wise patterns revealed that maximum fatalities in road accidents on the National Highways took place in Uttar Pradesh (13.6%) (6,137 out of 45,275 deaths) followed by Maharashtra (7.4%) (3,334 deaths), Karnataka (7.4%) (3,330 deaths), Rajasthan (7.3%) (3,289 deaths) and Bihar (6.8%) (3,061 deaths) during 2020.

Maximum number of accidents on State Highways in the country occurred in Tamil Nadu (15,471 cases). Maximum fatalities in road accidents on State Highways were reported in Uttar Pradesh (5,324 out of 33,824 deaths) which accounted for 15.7% of total deaths due to road accidents on State Highways, followed by Maharashtra (3,365) (9.9%) during 2020. Maximum fatalities on the Expressways were reported in Uttar Pradesh contributing 60.8% (711 out of 1,169) followed by Haryana (9.2%), West Bengal (6.1%), Assam (5.7%) and Maharashtra (5.1%) during 2020. [Table-1A.7]

LIST-1A.3
Road Accidental Deaths in every 100 km-2020
(Road Category wise)

Type of Road	2020				
	Length of Road (in KM*)	Road Accident Cases	Total Number of Deaths	Cases per 100 KM	Deaths per 100 KM
National Highways	126350 [#]	106933	45275	85	36
State Highways	186908 [#]	87716	33824	47	18
Other Roads	5900858 [#]	160147	54102	3	1
Total	6215797 [#]	354796	133201	6	2

- As per data provided by States/UTs.

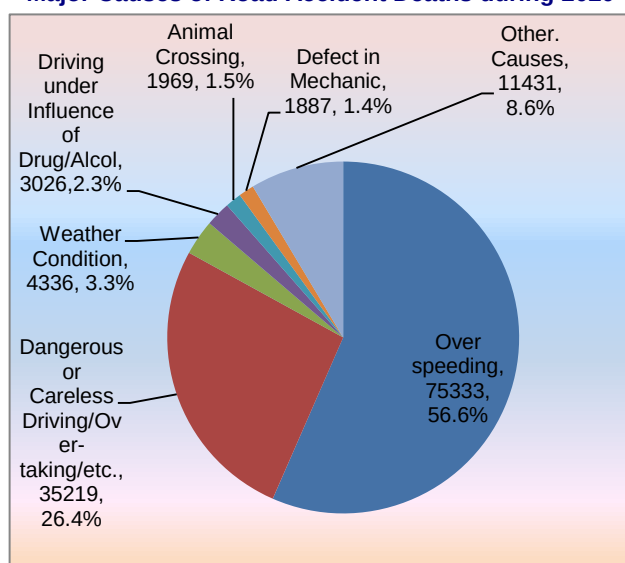
* Source: Road Accidents in India – 2019, Ministry of Road Transport & Highways.

figures of the previous year used due to non-availability of data.

Cause-wise distribution of road accidents (which also include unmanned railway crossing accidents) is presented in **Table-1A.8**. Cause-wise analysis of road accidents revealed that most of road accidents were due to over-speeding accounting for 60.6% of total accidents (2,15,159 out of 3,54,796 cases) which caused 75,333 deaths and injuries to 2,09,736 persons. Dangerous/careless driving or overtaking caused 86,268 accidents (24.3% of total accidents) which resulted in 35,219 deaths and injuries to 77,067 persons during 2020. 2.4% (8,664 out of 3,54,796 cases) of such accidents were due to poor weather condition. Driving under influence of drug/alcohol contributed 2.3% of total accidents which resulted in injuries to 6,974 persons & 3,026 deaths in the country.

FIGURE–1A.6

Major Causes of Road Accident Deaths during 2020



- As per data provided by States/UTs.

Cause - wise analysis of fatal road accidents revealed that 56.6% (75,333 out of 1,33,201 deaths) and 26.4% (35,219 out of 1,33,201 deaths) of fatalities in road accidents were due to over-speeding and dangerous/careless driving or overtaking respectively. Poor weather conditions, Animal crossing and mechanical defects in motor vehicles caused 3.3% (4,336 deaths) ,1.5% (1,969 deaths) and 1.4% (1,887 deaths) of total deaths due to road accidents respectively during 2020.

Large number of deaths in road accidents due to over- speeding were reported in Karnataka (10.4%) (7,809 out of 75,333 deaths) followed by Madhya Pradesh (9.4%) (7,086 out of 75,333 deaths). Dangerous/careless driving or overtaking caused maximum fatalities in Uttar Pradesh (10,171 out of 35,219) which accounted for 28.9% of total deaths followed by 10.8% (3,817) deaths in Maharashtra. Maximum fatalities due to driving under influence of drug/alcohol were reported in Uttar Pradesh (17.9%) followed by Telangana 11.3%) , Jharkhand (10.4%), Tamil Nadu (9.4%) and Madhya Pradesh (8.7%) during 2020 [**Table-1A.9**].

A total of 24 accidental deaths were reported at un-manned railways crossings. 83.3% of such deaths were reported in Tripura (20 out of 24 deaths) during 2020 [**Table-1A.9**].

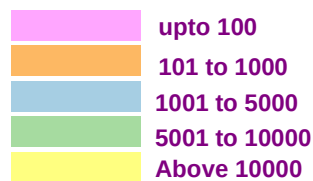
Place of occurrence - wise patterns of road accidents reveal that 59.6% of total accidents have occurred in rural areas (2,11,351) and 40.4% in urban areas (1,43,445) out of 3,54,796 total cases during 2020. Both in rural as well as urban area most of the accidents were reported at places near residential area. 31.3% (66,120 out of 2,11,351 cases) accidents in rural area and 32.6% (46,759 out of 1,43,445 cases) in urban area have taken place near residential area. 5.0% of road accidents in urban area took place at pedestrian crossing (7,154 out 1,43,445 cases). Besides, 8.0% (28,226 out of 3,54,796 cases) of road accidents in the country have taken place near schools, college or other educational institutions [**Table 1A.10**].

Uttar Pradesh followed by Bihar have reported 20.8% and 11.3% of total deaths due to road accidents near schools/college/other educational institutes in urban area respectively. Uttar Pradesh also reported highest fatalities due to road accidents at places near to residential area (urban area) accounting for 17.1% (2,581 out of 15,097) of total such deaths during 2020 [**Table-1A.11**].

FIGURE-1A.7
STATE/UT – WISE ROAD ACCIDENT DEATHS DURING 2020



Road Accident Deaths



- As per data provided by States/UTs.

Railway Accidents

A total of 13,018 cases of 'Railway Accidents' were reported, showing an decrease of 53.5% during the year 2020 over 2019 (27,987). 13,018 railways accidents caused injuries to 1,127 persons and 11,968 deaths during 2020 [Table-1A.1 & 1A.2].

Maximum railway accidents were reported in Maharashtra accounting for 20.7% (2,697 out of 13,018 cases) followed by Uttar Pradesh (12.0%) (1,560 cases). These two States have also reported highest fatalities in railways accidents, accounting for 16.1% (1,922 out of 11,968 deaths) and 13.0% (1,558 deaths) of total deaths in railways accidents respectively. 792 out of 1,127 persons injured in railways accidents were reported in Maharashtra alone during 2020 [Table 1A.2].

The month-wise distribution of 'Railway Accidents' shows that most of railway accidents were reported in the month of January (2,138 cases), contributing 16.4% of total railway accidents. Maharashtra (523 out of 2,138 cases) has reported maximum railways accidents in the month of January, accounting for 24.5% of total such accidents [Table-1A.5].

Most of railway accidents (2,155 out of 13,018) have taken place during 09:00 hrs to 12:00 hrs (Day), accounting for 16.6% of total railway accidents. 16.1% (2,092 cases) railways accidents were reported during '06:00 hrs to 9:00 hrs (Day)'. Maharashtra has reported maximum accidents during 09:00 hrs to 12:00 hrs (Day) and 18:00 hrs to 21:00 hrs (Night), accounting for 21.5% (463 cases) and 24.4% (431 cases) respectively [Table-1A.6].

State/UT - wise classification of railways accidents is presented in Table-1A.12 and Figure-1A.9. The analysis of classification of railway accidents revealed that incidents of 'Fall from Trains or Collision with People at Track' constituted majority of railway accidents (70.0%) (9,117 out of 13,018). State of Maharashtra has reported the majority of such cases, accounting for 27.9% (2,547 out of 9,117 cases) of total cases of fall from train or collision of trains with people at track. A total of 8,400 persons died due to either fall from trains

or collision of trains with people at tracks, accounting for 70.2% of total deaths in railway accidents (11,968 deaths).

State/UT - wise causes of railways accidents is presented in Table-1A.13. The causes include Fault of Driver, Sabotage, Signalmen's Error, Mechanical Failure and Other causes. Majority of States/UTs have furnished railways accidents under unclassified category 'Other Causes', a total of 12,440 out of 13,018 cases of railways accidents were furnished under 'Other Cause' (fall of persons from trains, persons coming under trains etc. come under this category). During 2020, a total of 23 cases of railways accidents occurred due to fault of driver. Mechanical defects (like poor design, track faults, bridge/tunnel collapse, etc.) caused 413, 51 and 29 railways accidents in Assam, Jharkhand and Uttar Pradesh respectively. In Jharkhand, a total of 18 persons died in railways accidents due to fault of driver. Mechanical defects (like poor design, track faults, bridge/tunnel collapse, etc.) led to loss of 539 lives in railways accidents during 2020.

Maximum railway crossing accidents were reported in Uttar Pradesh accounting for 37.5% (380 out of 1,014 cases) followed by Bihar (18.8%) (191 cases) and Madhya Pradesh (14.2%) (144 cases). These States have also reported highest fatalities in railway crossing accidents, accounting for 47.3% (561 out of 1,185 deaths), 16.1% (191 deaths) and 12.0% (142 deaths) respectively during 2020 [Table-1A.2].

Traffic Accidents in Cities

A total of 48,967 traffic accidents were reported in 53 cities during 2020. 48,967 traffic accidents caused injuries to 41,259 persons and 12,461 deaths. The maximum fatalities in traffic accidents was reported in Delhi City (1,568 deaths) followed by Chennai (872 deaths) and Jaipur (708 deaths) [Table-1A.2].

Road Accidents (47,829 cases) accounted for 97.7% of total traffic accidents in 53 mega cities during 2020. Chennai accounted for 9.2% (4,389 out of 47,829 cases) of total road accidents reported in 53 mega cities followed by Delhi City (8.4%) (4,028 cases) and

Bengaluru (6.8%) (3,233 cases). However, the large number of fatal road accidents were reported in Delhi City (1,151 deaths) followed by Chennai (872 deaths), accounting for 10.1% and 7.6% of total deaths due to road accidents in 53 mega cities respectively during 2020.

Cause-wise analysis of road accidents (including unmanned railway crossing) revealed that most of road accident deaths (including unmanned railway crossing) in 53 mega cities were due to over-speeding which accounted for 54.8% (6,260 out of 11,433 deaths) of total deaths due to road accidents during 2020. Dangerous or Careless driving/Over-taking etc. also caused 27.6% of total deaths due to road accidents (3,155 out of 11,433 deaths). Driving under influence of drug/alcohol had caused 4.0% (462 out of 11,433 deaths) of fatalities in road accidents. Among 53 mega cities, most of fatalities due to driving under influence of drug/alcohol were reported in Chennai (236 out of 462 deaths) [Table-1A.9].

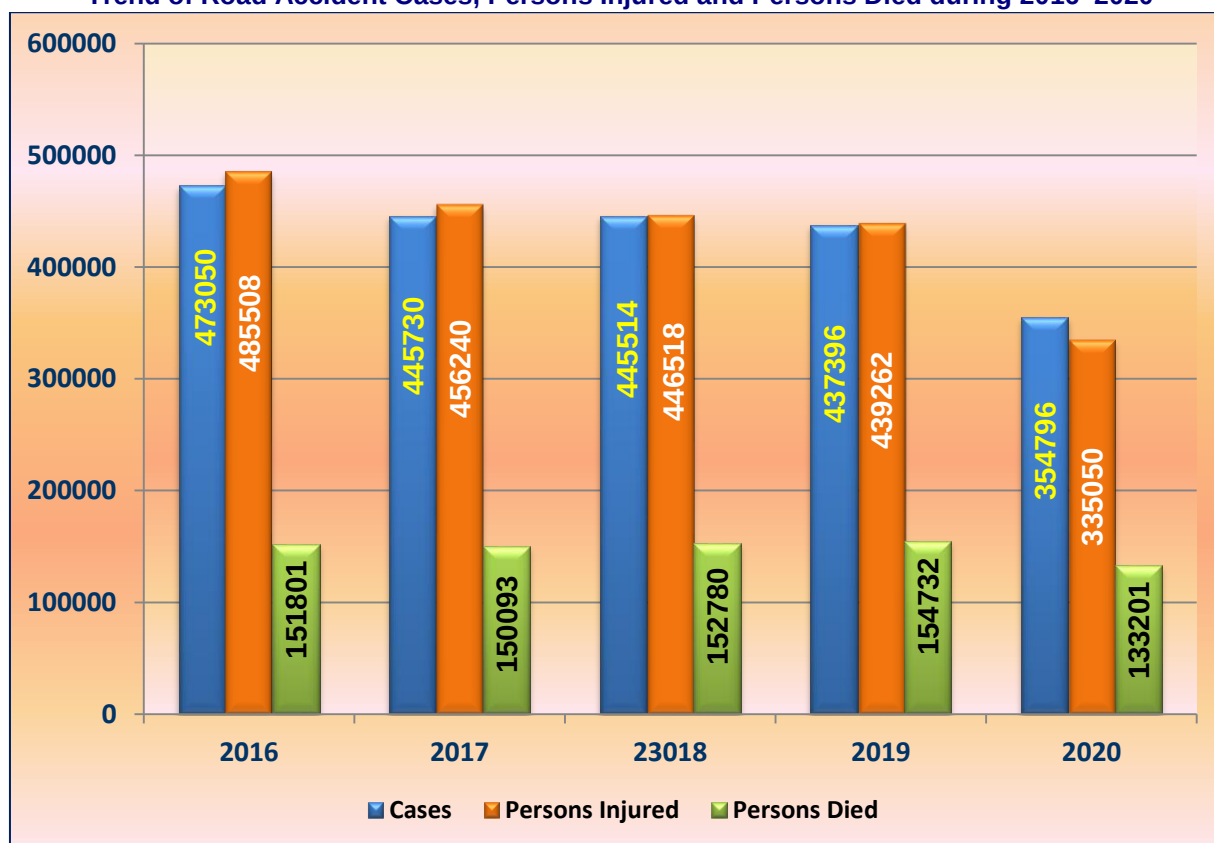
accidents reveals that most of fatalities due to road accidents have taken place near residential area, contributing 33.3% (3,802 out of 11,433 deaths) of deaths in road accidents in 53 mega cities, followed by 9.4% (1,074 out of 11,433 deaths) near factory/industrial area and 7.9% near schools/ college/other educational institutions (902 out of 11,433) deaths. Out of 53 mega cities, Chennai (151 deaths) followed by Ahmedabad (121 deaths) have reported maximum cases of road accidents at pedestrian crossing. [Table-1A.11]. As per road-wise classification of road accidents, 22.4% of total road accidents in 53 mega cities were reported at the National Highways. 28.6% of fatalities in road accidents were reported at the National Highways during 2020 [Table-1A.7].

A total of 999 railway accidents were reported in 53 mega cities where in Delhi City has reported maximum incidents by contributing 52.7% of total railway accidents during 2020.

Place of occurrence – wise deaths in road

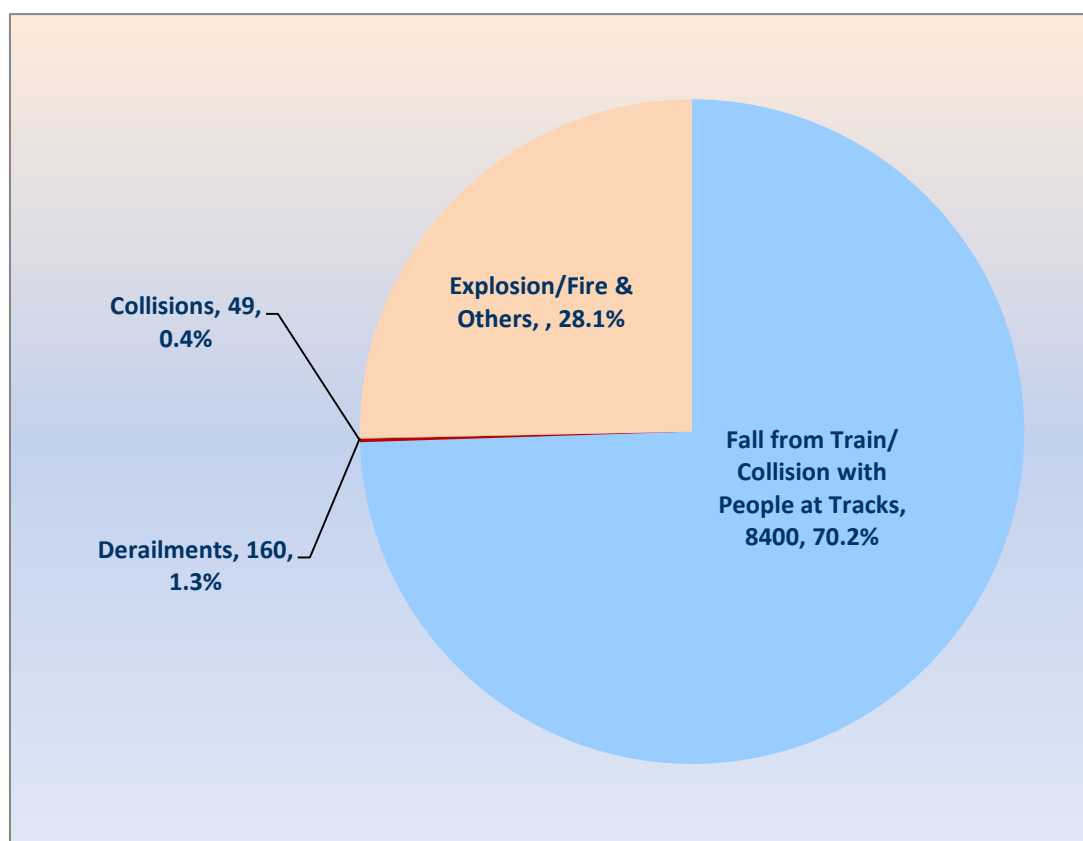
FIGURE–1A.8

Trend of Road Accident Cases, Persons Injured and Persons Died during 2016–2020



- As per data provided by States/UTs.

FIGURE-1A.9
Classification of Railway Accident Deaths during 2020



- As per data provided by States/UTs.
